Batterjee Medical College
For Science and Technology

Adrenal Gland Disorders

Internal Medicine Department

Knowledge

- Relate pathophysiological basics and aetiology of adrenal gland disorders
- Describe epidemiology, manifestations, complications and management of hyper and hypoadrenalism
- Describe causes, manifestations of hirsutism

Skills

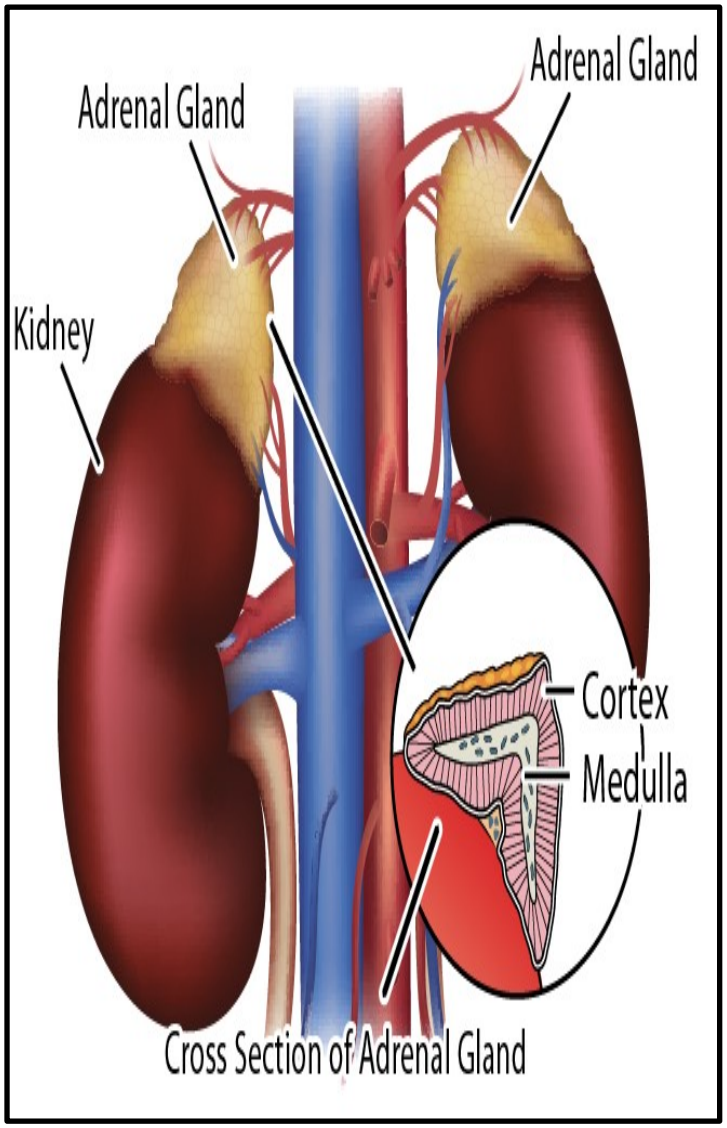
- Interpret clinical manifestations and investigations of adrenal gland disorders to formulate reasonable diagnosis.
- Apply evidence-based management plan for adrenal gland disorders
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

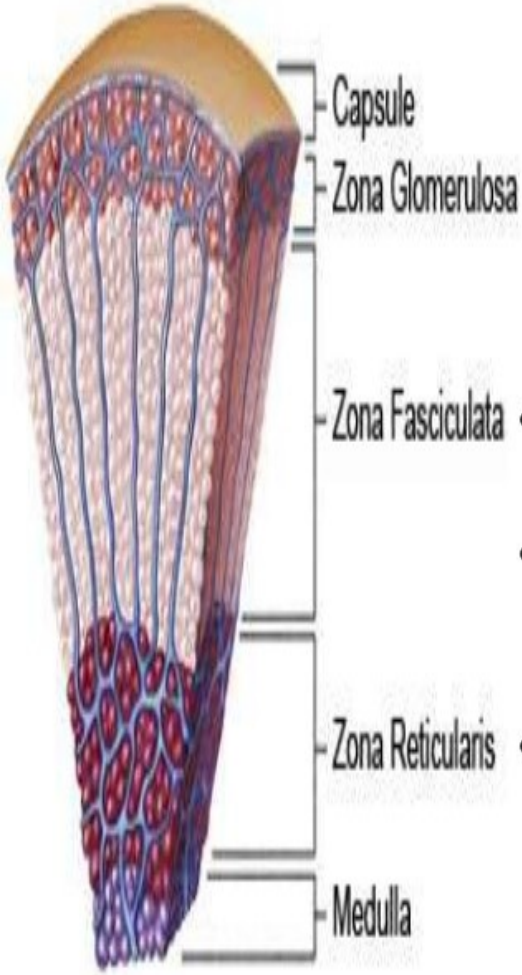
Main Subtopics

- Adrenal gland anatomy and physiology (histology, steroidogenesis, renin-angiotensin-aldosterone system)
- Primary hyperaldosteronism/Conn's syndrome (causes, clinical picture, investigations, treatment, differential diagnosis)
- Cushing syndrome (causes: pituitary, adrenal, ectopic, exogenous; clinical picture)
- Investigations of Cushing syndrome (screening tests, confirmatory tests, differentiation between causes, localization, differential diagnosis)
- Treatment of Cushing syndrome and Nelson's syndrome

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THE ADRENAL GLAND





- Histologically, adrenal cortex comprises *three* distinct layers:
  - *(i) outer zona glomerulosa*
  - *(ii) middle zona fasiculata (largest layer)*
  - *(iii) inner zona reticularis*
- Cells of *all three* zones secrete *corticosterone*
- Enzymes responsible for *aldosterone* production occur only in the *zona glomerulosa*
- Enzymatic mechanism for synthesizing *cortisol (hydrocortisone) and androgens* exists mainly in the *two inner zones*

ADRENAL GLAND

Covered with thick CT capsule from which trabeculae extend into parenchyma, carrying blood vessels & nerves



OUTER YELLOWISH CORTEX

- Steroid- secreting
- 90% of the gland by weight
- Mesodermal mesenchyme
- Controlled in part by anterior pituitary gland
- Regulate metabolism & maintain normal electrolyte balance

ZONA GLOMERULOSA

mineralocorticoids

ZONA FASICULATA

glucocorticoids

ZONA RETICULARIS

androgens

DARK INNER MEDULLA

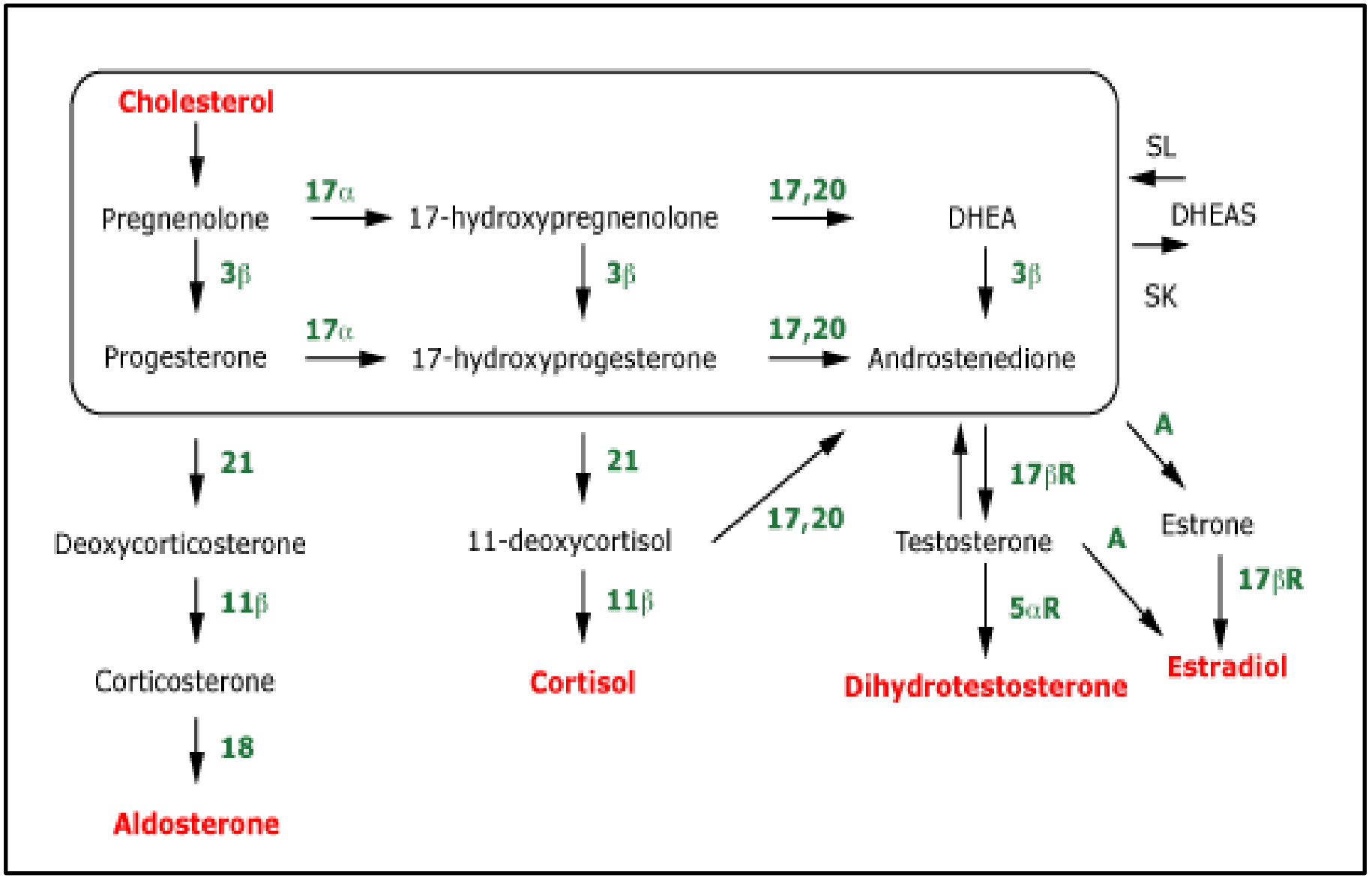
- Catecholamine-secreting
- Form center of the gland
- Neuroectoderm
- Richly innervated by preganglionic sympathetic fibers

CHROMAFFIN CELLS

catecholamines

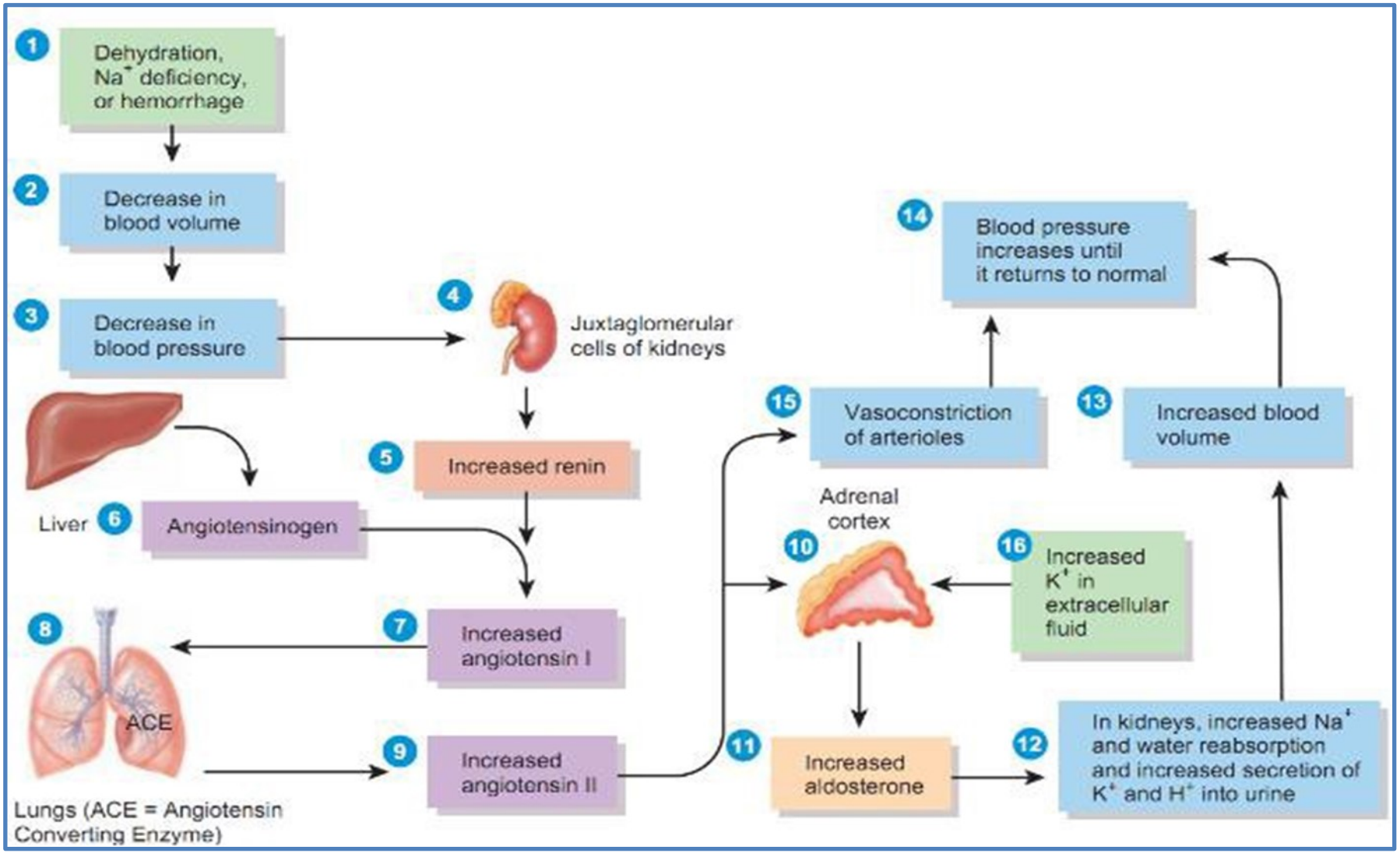
SYMPATHETIC GANGLION CELLS

Synthesis of Adrenal Hormones



Physiological consideration

1.Zone glomerulosa: →secretes aldosterone → Na &H2O retention.



2.Zona fasciculate: secretes glucocorticoids, mainly cortisol (hydrocortisone).

Metabolic effects of cortisol:

- On fat (in excess)→ mobilization & deposition in abnormal sites.
- On proteins → catabolic, (in excess)→ muscle wasting & osteoporosis.
- On carbohydrates:→↑ gluconeogenesis & ↓ glucose uptake by the muscle cells, (in excess)→ hyperglycemia.
- On H2O & electrolytes→Na & H2O retention & k execretion.
- On blood (bone marrow)→ ↑ RBCs &↓ eosinophils & lymphocytes.
- Androgenic.
- Anti-allergic (tend to – Ag-Ab reaction).
- Anti-inflammatory (& in excess ↓fibrous tissue formation & destroys elastic fibers).

3.Zona reticularis : Secretes sex hormones, mainly androgens & small amount of oestrogen & progesterone.

- The main androgen is dehydro-epiandrosterone which enhances protein anabolism & promotes the development of male 2ry sexual characters.
- The end products of androgens are ketosteroids (which are excreted urine).
- It is under the control of ACTH.

N.B: Sources of androgens :

- In ♂ → (adrenal, testes).
- In ♀ →(adrenal).

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CONN'S SYNDROME
(1ry hyperaldosteronism)

Causes

1.Adrenal aldosterone producing adenoma (60%).

2.Hyperplasia of zona glomerulosa (bilateral) (40%).

3.Aldosterone producing carcinoma (rare).

Clinical picture

1.Hypertension: may be severe, due to → Na retention.

2.Hypokalaemia: leading to:

- Apathy, paraesthesia.
- Arrhythmias as extrasystoles.
- Atony of intestine→ constipation & even paralytic ileus.
- Muscle weakness & even episodic paralysis
- Impaired glucose tolerance as K has a direct effect on Insulin release from B cells & may Influence Insulin action.

3.Alkalosis “metabolic”: Due to heavy loss of H+ in urine→ tetany.

4.NO oedema: due to….

- K diuresis → polyuria.
- Impaired tubular reabsorption of water (renal concentration defect) →Polyuria " due to prolonged Hypokalemia'.

- Aldosterone escape 9
  - Definition: Evasion of the Na^+-retaining effects of inappropriately elevated aldosterone levels in conditions such as primary hyperaldosteronism or congestive heart failure
  - Mechanism: sodium and water retention → volume expansion → secretion of atrial natriuretic peptide (ANP) and pressure natriuresis → compensatory diuresis → "escape" from edema formation and hypernatremia

Clinical pearl

In edematous disorders the aldosterone escape mechanism is impaired, resulting in worsening edema.

Investigations

1.Biochemical changes :

a. Hypokalaemia (N = 3.5 -- 5 mEq / L).

b.Hypernatraemia (N =140 mEq / L).

c. Alkalosis (↑ serum HC03 mEq / L) (N= 22-30 mEq /L).

2.Urinary changes:

a. Polyuria.

b.↑ K, ↓ Na.

c. ↑ aldosterone excretion(N = 12 -50 ug / 24 hs).

3.Measurement of plasma level of :

a. Renin:→↓ in lry hyperaldosteronism (& remains low on Na restriction).

- Plasma renin level is high in cases of 2ry hyperaldosteronism.
- You must stop ttt of HPN before measurement of renin.

b. Aldosterone : high (N =3 - 10 ng / dl).

4. U/S & adrenal C.T.

Confirmatory testing in hyperaldosteronism: to verify if production is nonsuppressible (i.e., not regulated by the RAAS). Options:

- **Oral sodium loading test:** Ensure high sodium intake for 3 days and collect 24-hour urine aldosterone on the last day. In primary hyperaldosteronism is highly likely if urinary aldosterone > 12 mcg/day
- **Saline infusion test:** Draw baseline laboratory studies (e.g., renin, aldesterone), infuse normal saline over 4 hours, and draw laboratory studies again. In primary hyperaldosteronism is very probable in patients with aldosterone levels > 10 ng/dL
- **Others:** fludrocortisone suppression test or captopril suppression test

➢ Interpretation

  - Aldosterone suppression after interventions: primary hyperaldosteronism unlikely. Consider other diagnoses.
  - No aldosterone suppression after interventions: primary hyperaldosteronism confirmed

Treatment :

1.Surgical removal of the adenoma (after correction of the electrolyte imbalance).

2.Medical ttt :

- In bilateral adrenal hyperplasia.
- Amiloride (40 mg / day) is preferable to spironolactone (which leads to gynaecomastia & impotence on chronic use).

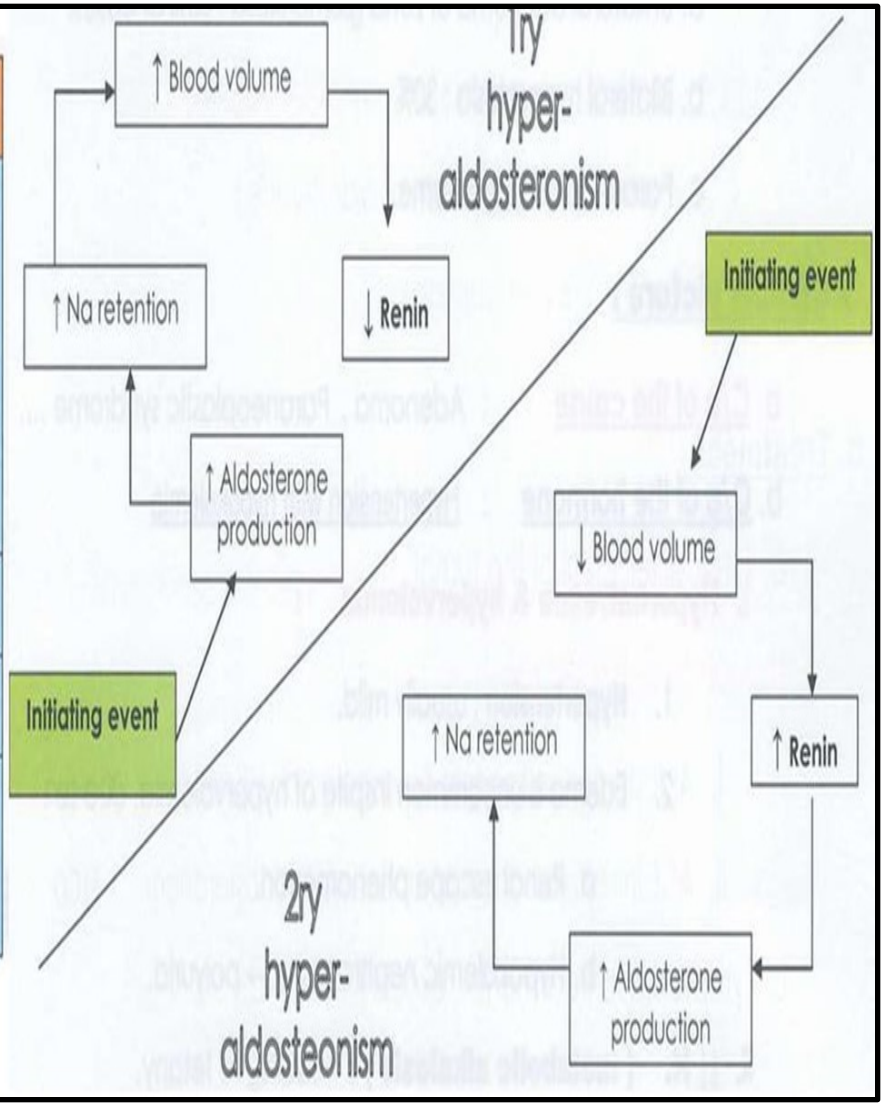
Differential diagnosis:

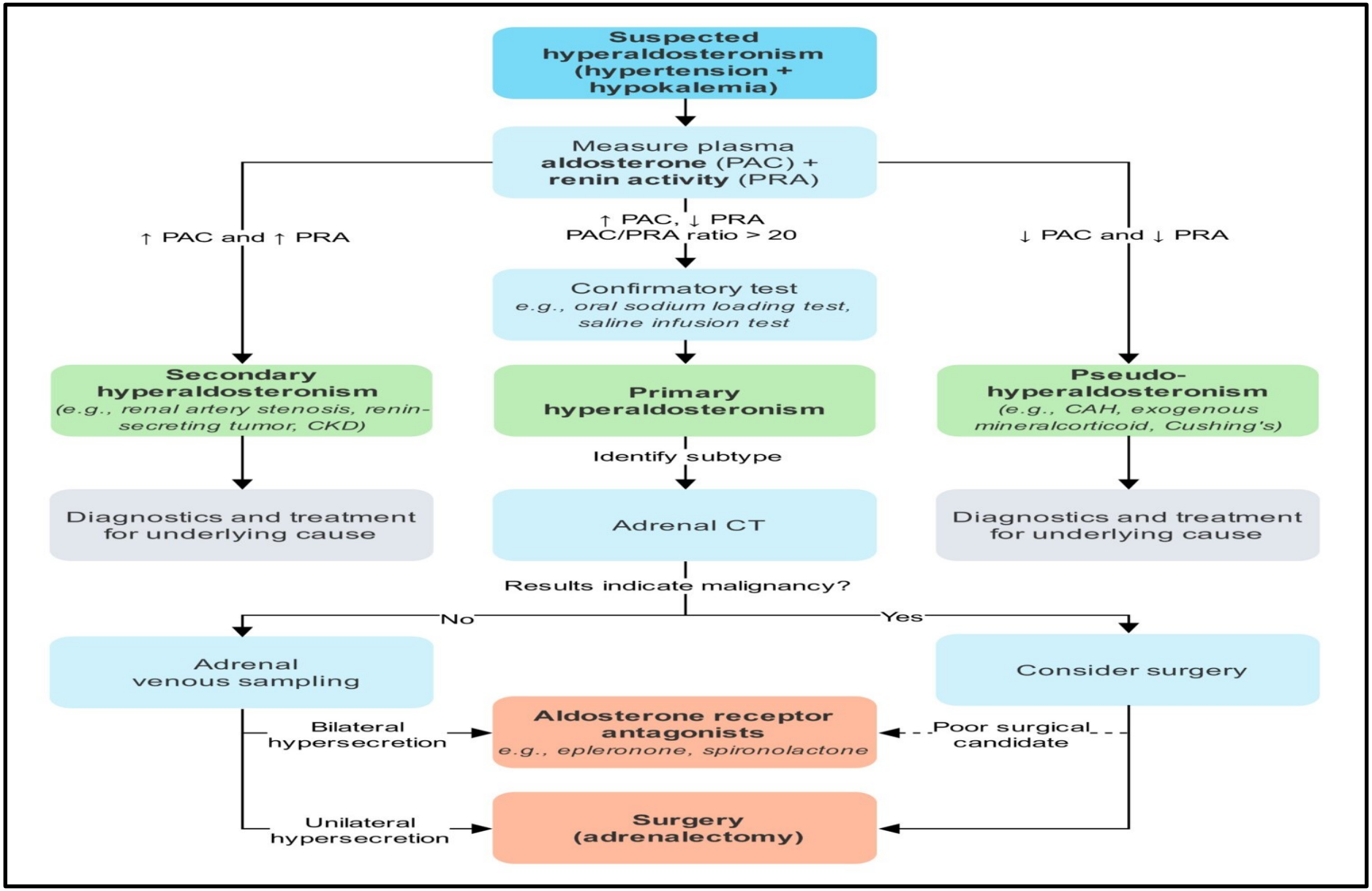
A. from hypertensive conditions associated with hypokalemia (Pseudohyperaldostronism):

- Hypertension ‘malignant or renal’ with 2ry hyperaldosteronism.
- Diuretics causing Hypokalaemia in hypertensive patient.
- Cushing syndrome, CAH & rarely Juxtaglomerular hyperplasia
- Intake of exegenous mineralocorticoids.
- K-losing nephropathies causing or associated with HPN e.g. bartter’s Syndrome, Liddle’s syndrome.

B. from other causes of hypokalemia & familial periodic paralysis.

	1ry hyperaldosteronism	2ry hyperaldosteronism
Source	- Adrenal Gland.	- Renal artery stenosis. - Heart failure. - Nephrotic syndrome. - Liver cirrhosis.
Aldosterone	+++	+
Renin	↓↓	↑↑
HTN	mild.	No HTN except in renal artery stenosis → sever.




Suspected hyperaldosteronism (hypertension + hypokalemia)
Measure plasma aldosterone (PAC) + renin activity (PRA)
↑ PAC and ↑ PRA
↑ PAC, ↓ PRA
PAC/PRA ratio > 20
↓ PAC and ↓ PRA
Confirmatory test
e.g., oral sodium loading test, saline infusion test
Secondary hyperaldosteronism
(e.g., renal artery stenosis, renin-secreting tumor, CKD)
Primary hyperaldosteronism
Pseudo-hyperaldosteronism
(e.g., CAH, exogenous mineralcorticoid, Cushing's)
Identify subtype
Diagnostics and treatment for underlying cause
Adrenal CT
Diagnostics and treatment for underlying cause
Results indicate malignancy?
No
Yes
Adrenal venous sampling
Consider surgery
Bilateral hypersecretion
Aldosterone receptor antagonists
e.g., epleronone, spironolactone
Poor surgical candidate
Unilateral hypersecretion
Surgery (adrenalectomy)

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CUSHING'S SYNDROME

Causes

A. Endogenous Cushing syndrome " rare, yearly incidence 1 /million ".

1. Pituitary Cushing syndrome (Cushing disease).

- The most common cause of endogenous Cushing (75 %).
- 80 % are caused by pituitary basophil micro—adenoma secreting excessive of ACTH " the remaining cases may be of hypothalamic origin "

2.Adrenal Cushing syndrome: (20 %): Caused by autonomous cortisol production from adenoma, carcinoma or hyperplasia of adrenals.

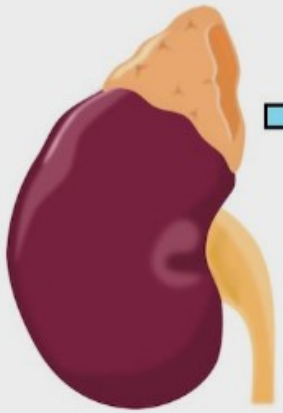
3.Ectopic Cushing syndrome . (autonomous ACTH production) e.g.

- Bronchogenic carcinoma, pulmonary carcinoid.
- Thymoma, ovarian or pancreatic carcinoma.

B. Exogenous Cushing syndrome (Cushingoid syndrome) produced by prolonged use of corticosteroids

Cushing's Syndrome vs Disease

Cushing's Syndrome

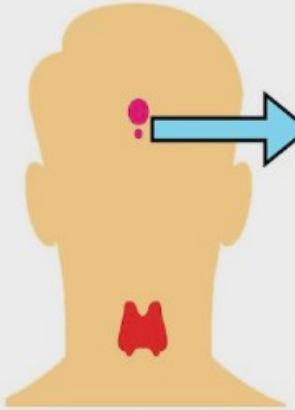


Cortisol

Cushing's Syndrome: An endocrine disorder in which cortisol levels are too high in the body

There are a number of underlying causes (adrenal tumor, pituitary tumor, corticosteroid use, etc.)

Cushing's Disease



Cortisol

Cushing's Disease: Occurs when Cushing's syndrome is caused by a pituitary tumor

Cushing's disease is a specific subtype of Cushing's syndrome caused by a pituitary tumor

Clinical picture

A. Abnormal deposition of fat in certain sites :

- In the face→ becomes rounded with bloated cheeks (Moon face).
- In the interscapular region → (Buffalo hump).
- In the breasts & abdomen (Trunkal obesity), but the buttocks are hollow with thin limbs (lemon on match sticks).







B. Disturbances in protein metabolism:

- ↑ Protein catabolism →muscle wasting & weakness even proximal myopatny).
- Osteoporosis :
  - o Kyphosis, shortening
  - o Pathological fractures.
- Bruises & purpura (due to ↓ support of blood vessels).
- Delayed healing of wounds.
- Stria rubra : Due to rupture of weakened s c. collagen fibers (together with fat mobilization) →purplish lines in the skin around the breasts, abdomen & thighs.



C. Disturbances in carbohydrate metabolism :

Hyperglycaemia & may end in D.M. (in 15 % of cases) which is somewhat insulin resistant (steroid diabetes).

D. Disturbances in fluid & electrolytes balance :

- Na retention → hypertension.
- Hypokalaemia → polyuria, alkalosis.

E. General manifestations :

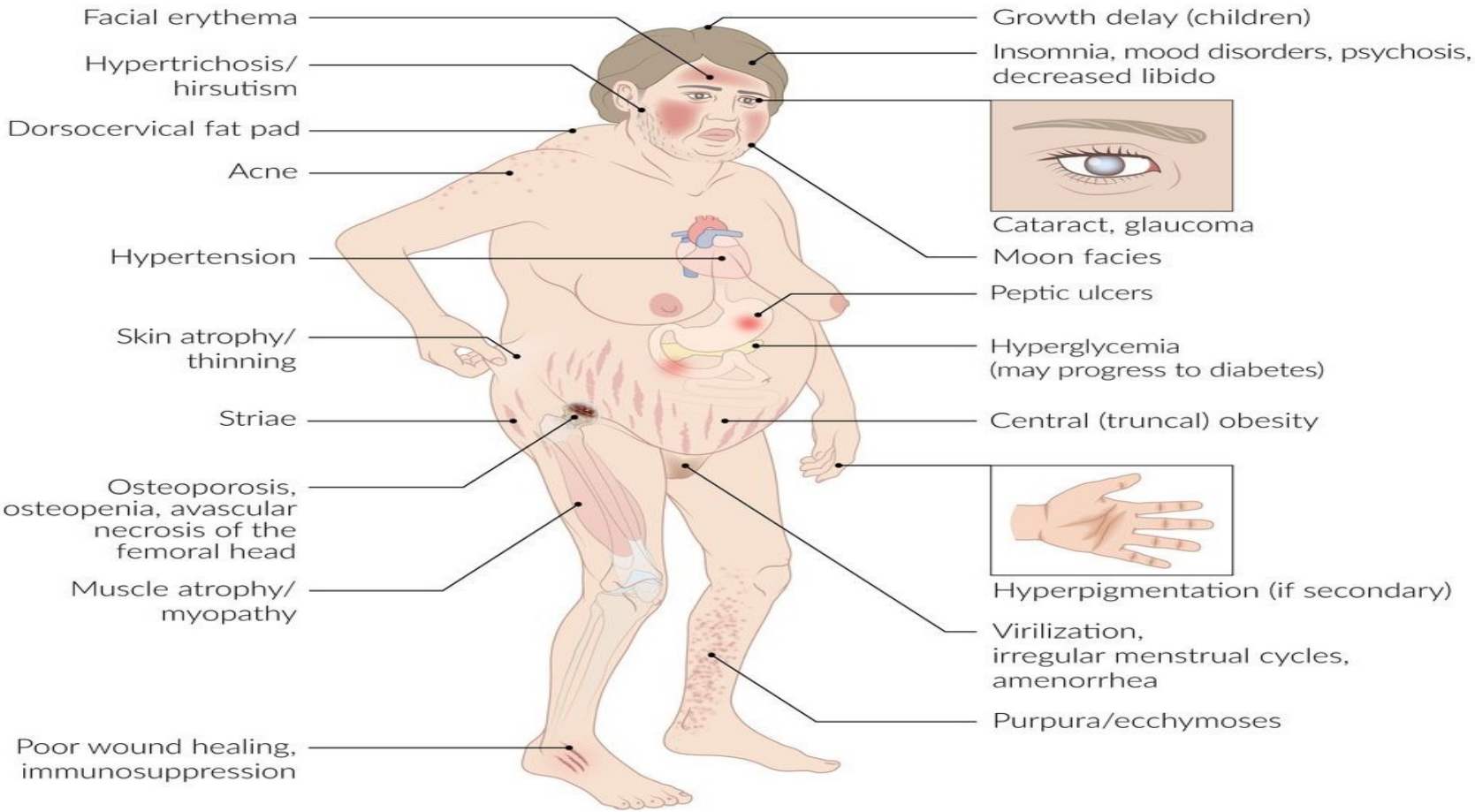
- Plethoric face due to polycythemia.
- Psychiatric disturbances: common depression.
- Obesity & stunted growth may be the main presenting feature in childhood.

F. Sexual manifestations :

- *in females :*Excess adrenalAndrogen
  - o Amenorrhea
  - o Acne
  - o Hirsutism.
- *in males :*
  - o ↓ Sexual desire.
  - o impotence.

G. features suggesting the aetiology :

- Evidence of Virilization (is most common) → adrenal carcinoma.
- Hypokalaemic alkalosis, myopathy & hyperpigmentation (occur most often in) → Ectopic Cushing syndrome (the distinguishing clinical features of hypercortisolism are often absent).
- Age & sex:
  - o Children (More common in) → adrenal carcinoma.
  - o Adult ♂ → Ectopic Cushing. “esp. smokers”.
  - o ♀ ‘in childbearing age’ → pituitary Cushing.



Student pearl

"**CUSHINGOID**" is the acronym for side effects of corticosteroids: Cataracts, Ulcers, Striae/Skin thinning, Hypertension/Hirsutism/Hyperglycemia, Infections, Necrosis (of the femoral head), Glucose elevation, Osteoporosis/Obesity, Immunosuppression, Depression/Diabetes.

Investigations

1.investigations which suggest Cushing syndrome :

a. Hypokalemic alkalosis. Hypernatremia.

b. Blood picture → ↑ RBCs, ↓ lymphocytes & eosinophils.

c. Impaired glucose tolerance.

d. X-ray → osteoporosis & unsuspected fractures

2.Investigations which confirm Cushing syndrome :

a. Plasma cortisol level : (N = 5 — 20 ug/dL)" at 8 AM "

- The earliest indication of ↑ cortisol secretion is loss of the diurnal
- Variation (circadian rhythm) "normally, plasma cortisol level at 4 pm. is < 50 % of that at 8 am."
- Then, there is persistent elevation.

b.Urinary steroid excretion : → ↑ '24 hours collection'

- Hydroxy corticosteroids (17 oH Cs).
- Urinary free cortisol excretion (most reliable index of hypercortisolism).
- Overnight dexamethasone suppression test: Dexamethasone 1 mg is given orally at night (I 1 pm) & plasma is obtained at 8 am the following morning → cortisol level < 7 ug/dl exclude hypercortisolism.

3.Investigations to differentiate between the 3 forms of Cushing syndrome :

a. Plasma ACTH:

- Very high → in Ectopic Cushing.
- High→in pituitary Cushing.
- low or absent → in adrenal Cushing.

b. High dose dexamethasone suppression test :

2 mg / 6 hours for 2 days is given orally.

- In pituitary Cushing → 50 % ↓ of 24 hours excretion of 17 OHCs or the plasma cortisol.
- In adrenal or Ectopic Cushing → no effect

4. Localization procedures :

- Abdominal U/ S & CT, Adrenal, retrograde venograpy for adrenal tumours.
- Sellar X-ray detects 10 -15 % of pituitary tumours & C.T. detects 85 % of microadenomas (MRI is better).
- Chest x-ray →for bronchial carcinoma.

Clinical pearl

Abdominal CT or MRI in a patient with Cushing disease will show bilateral hyperplasia of both the zona fasciculata and zona reticularis.

Differential diagnosis :

- Exogenous obesity, essential HPN & D.M. occurring separately or combined:

  In such cases : - Obesity is generalized.

  - Plasma cortisol & urinary end products →normal.

- Other causes of obesity: e.g. Myxoedema, Frohlich's synd.. etc
- Other causes of Hirsutism. (later)
- Other causes of osteoporosis. (later)
- Women on oral oestrogen - containing C.C.P.s may become overweight & may have higher levels of plasma cortisol (due to ↑ cortisol binding globulin).
- Patients suffering from LCF may develop Cushing - like manifestations due to impaired cortisol metabolism. In such cases, plasma cortisol level is raised but, manifestations of LCF are prominent.

N.B: Pseudo-Cushing syndrome: Chronic alcoholism, severe illness, depression & obesity & may be associated with mild ↑ in cortisol levels & absent diurnal variation

Treatment

1.Pituitary Cushing:

a) Surgery:

- Trans - sphenoidal hypophysectorny.
- Remission (85 %)
- Complications …..

b) irradiation e .g. Proton beams, alpha particles

- After ttt → cortisone for life

c) Medical:

- Drugs that ↓↓ ACTH release:
  - o Bromocriptine.
  - o Cyprohepatidine(hypothalamic seretonine antagonist).
  - o Na valproate [GABA transaminase inhibitor].
- Drugs that ↓↓ steroidogenesis

2.adrenal Cushing:

- Adrenal adenoma : → removal.
- Adrenal carcinoma →removal

o Preoperative: with enzyme inhibitor ‘metyrapone’ 11 deoxycortisol → metyrapone→ cortisol.

o Post-operative: Give mitotane” for residual disease" [adrenolytic agent].

3.Ectopic Cushing :

a) Surgical removal of the tumour. or

b) Metyrapone ± Mitotane.

Clinical pearl

Patients who develop adrenal insufficiency after surgery require lifelong glucocorticoid replacement therapy.

Student pearl

Enzyme inhibitors (e.g., metyrapone, ketoconazole) **suppress cortisol synthesis**, while glucocorticoid antagonists block the action of cortisol in peripheral tissues.

Nelson's Syndrome

- Bilateral adrenalectomy as treatment of Cushing's syndrome due to bilateral adrenal hyperplasia from an ACTH-producing pituitary tumor
- Hyperpigmentation
- Amenorrhea
- Expansion of pituitary tumor with invasive potential

- It is due to continuous positive feedback.
- It can be minimized by preoperative irradiation to pituitary gland

Case scenario :

A 32-year-old woman comes to the physician because of weight gain, generalized weakness, and irregular menstrual cycles for the past 16 months. She began having symptoms of insomnia and depression 10 months ago. More recently, she has been having difficulties rising from a chair. She has a 2-year history of hypertension. Current medications include citalopram and hydrochlorothiazide. She is 168 cm (5 ft 6 in) tall and weighs 100 kg (220 lb). BMI is 36 kg/m^2. Her pulse is 92/min, respirations are 18/min, and blood pressure is 134/76 mm Hg. She appears tired and has a full, plethoric face and central obesity. Examination of the skin shows violaceous linear striations on her lower abdomen. Two midnight serum cortisol studies show measurements of 288 µg/L and 253 µg/L (N < 90); a 24-hour urinary cortisol measurement was 395 µg (N < 300). Upon follow-up laboratory examination, the patient's serum ACTH levels were also elevated at 136 pg/mL (N = 7–50).

Which of the following is the most appropriate next step in evaluation?

A. MRI of the head with contrast
B. Measure ACTH levels in inferior petrosal sinuses
C. High-dose dexamethasone suppression test
D. Low-dose dexamethasone suppression test
E. CT scan of the abdomen with contrast
F. Administer ketoconazole
G. Bilateral adrenalectomy

- A 42-year-old man comes to the physician for a health maintenance examination. He has had generalized fatigue and muscle ache since his previous visit 6 months ago. He has hypertension and gastroesophageal reflux disease. Current medications include enalapril and omeprazole. His temperature is 37.1°C (98.1°F), pulse is 88/min, and blood pressure is 150/102 mm Hg. Physical examination shows no abnormalities. Serum studies show:
  - Na+ 143 mEq/L K+ 2.8 mEq/L Cl- 100 mEq/L
  - Urea nitrogen 14 mg/dL Creatinine 1 mg/dL
- Enalapril is discontinued and the patient is started on oral potassium chloride and hydralazine therapy. Two weeks later, his plasma aldosterone concentration is 35 ng/dL (N = 3.6–24.0) and plasma renin activity is 0.4 ng/mL/h (N = 0.3–4.2). An oral sodium loading test over 3 days fails to reduce aldosterone.

Which of the following is the most appropriate next step in management?

a) Adrenal venous sampling
b) Spironolactone therapy
c) Fludrocortisone suppression test
d) Adrenal CT scan
e) Adrenalectomy
f) MRI of the abdomen

Summary



References

1. AMBOSS
2. Step-up to Medicine, 6th edition (ISBN-13: 978-1975192730)
3. Davidson's Principles & Practice of Medicine, 25th edition (ISBN-13: 978-0443287626)
4. Harrison's Principles of Internal Medicine, 22nd edition (ISBN-13: 978-1265979317)
5. Current Medical Diagnosis & Treatment, 65th edition (ISBN-13: 978-1260469868)
6. Kumar and Clark's Clinical Medicine, 11th edition (ISBN-13: 978-0443128288)

Thank You